

Reorder Level Calculation

The reorder level is defined as the sum of the safety stock plus the requirement forecast within the replenishment lead time (see formula (17)).

Example

A forecast was carried out on a monthly basis. A month has 30 days in the case of external procurement.

Safety stock		100
Forecast	1st subsequent period	200
	2nd subsequent period	300
	3rd subsequent period	400
Replenishment lead time	40 days	0/30
		30/30 + 10/30

(an entire monthly requirement + a part of the following month)

Reorder level = $100 + 30/30 * 200 + 10/30 * 300 = 400$

Safety Stock Calculation

The safety stock depends on the service level that you specified in the MRP II view of the material master record and on the accuracy of the forecast. The more accurate the forecast, the smaller your safety stock can be.

The following figure shows that, without safety stock, customer demand can be satisfied by 50%. It also shows that it is almost impossible to satisfy customer demand 100% of the time. Factor R describes the relationship between forecast accuracy and service level (SL).

If replenishment lead time is greater than the forecast period by factor W, then the mean absolute deviation (MAD) is recalculated for this period (formula 17).

If the material is produced in-house, the delivery time is: opening period + in-house production time + goods receipt processing time. It is expressed in workdays. The forecast period is taken from the material master record and is also expressed in workdays.

If the material is procured externally, the delivery time is: goods receipt processing time + planned delivery time + goods receipt processing time. It is expressed in calendar days. The forecast period is taken from the material master record and is also expressed in calendar days.

i

The safety stock calculation may lead to rounding effects.

Minimum Safety Stock

You can specify a minimum safety stock. If the result of the safety stock calculation is lower than this limit, the safety stock is then automatically set to this minimum value. You enter the minimum safety stock in the material master record (the MRP 2 screen).

Evaluating Forecast Accuracy

Every forecast should provide a basis for a decision in some form. So that changes in a time series pattern can be recognized early, the following parameters are calculated in the SAP system:

- Error total
- Mean absolute deviation (MAD)
- Tracking signal
- Theil coefficient

Error Total

$$(12) \quad ET = \sum_{t=1}^n [V(t) - P(t)]$$

Mean Absolute Deviation for Initialization

$$(13) \quad MAD = \frac{1}{n} \sum_{t=1}^n |P(t) - V(t)|$$

n = Number of periods for initialization

Mean Absolute Deviation for the Ex-Post Forecast

$$(14) \quad MAD(t) = (1 - \delta) * MAD(t-1) + \delta * |V(t) - P(t)|$$

Tracking Signal

$$(15) \quad TS = \left| \frac{ET}{MAD} \right|$$

Theil coefficient for the Ex-Post Forecast

$$(16) \quad U = \sqrt{\frac{\sum_{t=1}^T (P(t) - V(t))^2}{\sum_{t=1}^T (V(t) - V(t-1))^2}}$$

Key

$P(t)$ = Forecast value

$V(t)$ = Actual value

ET = Error total

TS = Tracking signal

4. Stock conversion Lead-time (or Quality Inspection lead time)

A delay in any or all of the above can have effect on the entire replenishment process and the stock. A buffer stock must be designed to take into account the above coverage. Again the determination of your safety stock depends on the accuracy of your forecast. Higher your accuracy, lower your safety stock. This relationship between forecast accuracy and service level

is denoted by factor R. This also takes into account that the customer demand cannot be always satisfied 100% of the time.

Hence what we have is:

R = Relationship between forecast accuracy and service level (Service Factor)

W = Delivery time (in days) / Forecast Period (in days)

MAD = Mean absolute deviation (parameter for forecast accuracy)

Now If replenishment lead time is greater than the forecast period by factor W then:

Safety Stock = $R \times \text{Sq.rt. } W \times \text{MAD}$

Else

Safety Stock = $R \times W \times \text{MAD}$

Now in SAP If the material is produced in-house, the delivery time is: opening period + in-house production time + goods receipt processing time. It is expressed in workdays. The forecast period is taken from the material master record and is also expressed in workdays.

Calculation of the Standard Deviation and MAD

Use

The system calculates the standard deviation and the mean absolute deviation (MAD). This standard deviation is also used by other planning services in **Service Parts Planning**, such as inventory planning.

Features

The calculation of the standard deviation and the MAD is part of the forecast model itself in the following forecast models:

- [intermittent forecast model](#)
- [seasonal trend model with FPG](#)

For all other models the system uses the [historical forecast](#) to calculate the standard deviation and the MAD.

The system calculates the **standard deviation** as follows:

$$\text{Std Dev.} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (\text{Fcst}_{i(\text{from } j - \text{RLT})} - \text{DemHist}_i)^2}$$

https://help.sap.com/saphelp_SCM700_ehp02/helpdata/en/b0/2bc95360267614e10000000a174cb4/content.htm?no_cache=true

The system calculates the **MAD** as follows:

1. The system initializes the first three of the historical periods you defined for calculating the MAD

The system proceeds as follows:

$$\begin{aligned} \text{MAD Base Value} = & \text{Weighting factor1} * | \text{DemHist1} - \text{Fcst1} | + \\ & \text{Weighting factor2} * | \text{DemHist2} - \text{Fcst2} | + \\ & \text{Weighting factor3} * | \text{DemHist3} - \text{Fcst3} | \end{aligned}$$

$$\text{Std Dev.} = \sqrt{\frac{1}{n-1} * \sum_{i=1}^n (\text{Fcst}_{i(\text{from}_j - \text{RLT})} - \text{DemHist}_i)^2}$$

MAD Base Value = Weighting factor1 * | DemHist1 – Fcst1| +
 Weighting factor2 * | DemHist2 – Fcst2| +
 Weighting factor3 * | DemHist3 – Fcst3|

$$\text{MAD}(i) = \alpha * |V(i) - \text{Fcst}_{i(\text{from}_j - \text{RLT})}| + (1 - \alpha) * \text{MAD}(i-1), i \geq 4$$

2. The system calculates the MAD according to the following formula:

$$MAD(i) = \alpha * |V(i) - Fcs_{i-(i-n-j-RLT)}| + (1 - \alpha) * MAD(i-1), i \geq 4$$

i represents the current period.

You can specify **alpha** from the **SAP Easy Access** menu under ► **Advanced Planning and Optimization** > **Service Parts Planning (SPP)** > **Planning** > **Forecast** > **Forecast Profile** ▾ on the **Model Parameter** tab page in the **Alpha Factor** field. In the DMA model you specify alpha on the **Model Parameter** tab page in the **MAD Smoothing Factor for Demand in DMA Model** field.

i-RLT represents the period, in which the forecast for period i was created. You can specify this value in the forecast profile on the **General** tab page in the **Offset Forecast Creation** field.

n is the number of periods that you defined in the forecast profile on the **Model Parameter** tab page.

The following table gives you an overview of which parameters you must define for which forecast model:

Use

The system calculates the standard deviation and the mean absolute deviation (MAD). This standard deviation is also used by other planning services in **Service Parts Planning** , such as inventory planning.

Features

The calculation of the standard deviation and the MAD is part of the forecast model itself in the following forecast models:

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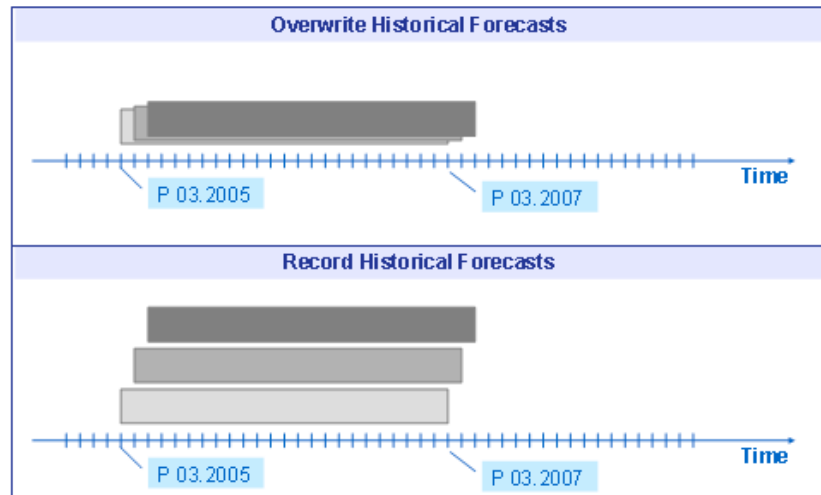
Historical Forecasts

Use

The system calculates the forecast for the number of periods that you specified in the forecast profile in the **Forecast Periods** field, and saves the result. If the system then recalculates the standard deviation one period later, it in turn calculates the forecast for this number of periods and saves the result. However, in doing so, the system overwrites the forecast for all periods apart from the first period of the previous calculation. This means that you can no longer access the forecast results previously calculated.

However, if a location has a replenishment lead time of more than one period, for example, you may also need the results of the previous forecast calculation to plan exactly. The system therefore records all calculated forecast results as historical forecasts in the above operation.

The following figure shows the overwrite operation and the recording of historical forecasts:



Let me try and explain to you through a mix of functional procedures and how SAP calculates the Safety Stock:

In normal materials management there are basically 2 basic inventory management systems, the P system and the Q system.

P system is a system wherein replenishment is done keeping the quantity constant. The period becomes the variant. In other words you fix the quantity you want the stock to dip, to trigger a requirement. As soon as the stock level is reached you replenish the stock. During a lean period the time taken to reach the level will be longer and during an active period the time taken to reach that stock could drop. This normally relates with your consumption based planning.

Q system is a system wherein replenishment is done keeping the period constant. The quantity becomes the variant. In other words you will check for the level of stock at fixed time intervals (daily, weekly, monthly etc.) compare it with the requirements on that day and trigger the procurement process for replenishment. This normally relates with your MRP.

Now there are 4 more factors that could affect the idealistic procurement pattern:

1. Ordering Lead-time.
2. Manufacturing Lead-time
3. Transporting Lead-time
4. Stock conversion Lead-time (or Quality Inspection lead time)

A delay in any or all of the above can have effect on the entire replenishment process and the stock. A buffer stock must be designed to take into account the above coverage. Again the determination of your safety stock depends on the accuracy of your forecast. Higher your accuracy, lower your safety stock. This relationship between forecast accuracy and service level

R factor for safety stock calculation

142 Views

Hi!

When computing safety stock for consumption-based planning, SAP uses in the SS formula a R factor that is the relationship between the service level and the forecast accuracy. The formula for calculating R is the following: (R factor = $z * 1,25$) , where Z is the Norm.Inv of the normal distribution with mean equal to zero and standard deviation equal to 1 and probability equal to the chosen service level.

Can anyone tell me what is the meaning of 1,25 ? Is there any statistical meaning to this constant?



AMIT K

Jun 30, 2009 at 06:40 AM

https://www.tutorialspoint.com/sap_scm/index.htm

According to Rupesh logic

R & MAD Values are based on historic data. These values are averaged out. If the Sigma is equal to 1.25 multiplied by the MAD.

Example

If the service level is 90%, the lead time is 4 days, and the mean absolute deviation from the forecast is 1000, the system calculates the safety stock as follows:

$$1.60 * 2 * 1250 = 4000$$

<https://blog.arkieva.com/relationship-between-mad-standard-deviation/>

YOUR CALCULATION SHOULD BE

<https://www.estepsoftware.com/papers/madsquare.pdf>

Service Level = 99.5%

<https://mathworld.wolfram.com/MeanDeviation.html>

Planned deliv time = 2 days

$$MAD = 447$$

$$R \times \sqrt{W} \times MAD \times 1.25(\text{Sigma}) = \text{Safety Stock}$$

$$2.571 \times 447 \times 1.25 = 1435.9875$$



Samunder Singh

Dec 30, 2008 at 07:25 AM

Dear,

Check These Inputs

On the DRP screen, select Planning ® Sales planning ® Safety stock to calculates the safety stock for the distribution center as follows:

Safety stock = ZF (safety factor) mutiplied by SQRLT (square root of the lead time) multiplied by the Sigma.

Where

The safety factor (ZF) is based on the service level you define in the material master record. For example, if the service level is 90%, the safety factor is approximately 1.60.

The lead time (LT) is equal to the sum of the planned delivery time and the goods receipt processing time, both of which you define in MRP 2 of the material master record.

The mean absolute deviation (MAD) is the difference between the actual and the forecasted demand.

The Sigma is equal to 1.25 multiplied by the MAD.

Example

If the service level is 90%, the lead time is 4 days, and the mean absolute deviation from the forecast is 1000, the system calculates the safety stock as follows:

$$1.60 * 2 * 1250 = 4000$$

$$(ZF * SQRLT * \sigma = \text{safety stock})$$

Ratio of MAD to The Forecast Error Standard Deviation

- When forecasts estimate the mean demand and forecast errors are normally distributed

$MAD = \int_{-\infty}^{\infty} |y - \mu| f(y) dy$, where $f(y)$ is the pdf of Normal distribution

Let us define a normalized variable by $t = (y - \mu) / \sigma$. Then $t \sim Normal(0,1)$

$$MAD = \int_{-\infty}^{\infty} |t\sigma| \phi(t) dt = \sigma \int_{-\infty}^{\infty} |t| \phi(t) dt, \text{ where } \phi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$$

$$MAD = 2\sigma \int_0^{\infty} t \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \text{ since } \phi(t) \text{ is symmetrical}$$

$$MAD = 2\sigma \frac{1}{\sqrt{2\pi}} = \frac{\sqrt{2}}{\sqrt{\pi}} \sigma \text{ since } \int_0^{\infty} t e^{-t^2/2} dt = 1$$

$$MAD \cong 0.8\sigma \text{ and } \sigma \cong 1.25MAD$$

Typically: **$CV = \sigma / \text{average value}$** , where σ is a standard deviation.

Then you put **$\sigma = 1.25 \text{ MAD}$** in the formula above and receive **$CV = 1.25 \text{ MAD} / \text{average value}$**

Below is a simple explanation why in case of normal distribution you can use equation **$\sigma = 1.25 \text{ MAD}$** .

Safety Stock with EOQ (Economic Order Quantity)

Economic order quantity (EOQ) is the ideal amount of stock a business should purchase to minimize inventory costs. It's useful when a company wants to minimize costs such as ordering, transportation and storage. The formula is written out as:

$$\text{EOQ} = \sqrt{DS/H}$$

"D" represents the demand for stock in a given period, "S" is the costs of these orders, and "H" represents the **holding costs** per item within the period.

Reorder Point Formula

This calculation helps companies determine the inventory level that would require dipping into safety stock:

$$\text{Reorder Point} = (\text{Average stock depletion in given period} \times \text{Average lead time}) + \text{Available safety stock}$$

Inventory Position Safety Stock Formula

This formula helps companies monitor net inventory, which is stock on hand minus any backorders:

$$IP = \text{Inventory on hand} - \text{Backorders} + \text{Inventory currently on order}$$

Economic Order Quantity (EOQ), also known as the Wilson formula, is a calculation used to determine the least costly number of units to order. The aim is to minimize the cost of ordering and holding stock, while still meeting demand and service level requirements.

The cost of ordering products is made up of the cost of placing your order, delivery, and transportation costs, and the cost of receiving the order. Holding costs consist of the financial costs of paying for stock in advance, warehousing and storage costs, and depreciation costs.

EOQ = economic order quantity in units

Q = estimated annual quantity used in units (can be found in the annual purchases budget)

O = estimated cost of placing one order

C = estimated cost to carry one unit in stock for one year

Ultimately the aim of EOQ is to have as little stock as possible while still being able to fulfill service demands. Clearly knowing what your safety stock requirements are is critical to achieving this fine balance.

Total quantity of a material or good required each day over a fixed period

$$\text{AVERAGE DEMAND} = \frac{\text{TOTAL USAGE OF ITEM FOR A SPECIFIC PERIOD}}{\text{NO. OF DAYS IN THAT PERIOD}}$$

EXAMPLE:

$$= \frac{\text{TOTAL USAGE OF ITEM FOR 1 CALENDAR MONTH}}{\text{NO. OF DAYS IN THAT MONTH}}$$



FUTURE DEMAND = AVERAGE DEMAND + DEMAND CREATED BY THE LARGE ORDER

EXAMPLE:

*If you manufacture car transmissions
and have received a large order*

= AVERAGE DEMAND + DEMAND CREATED BY THE LARGE ORDER



DEMAND VARIABILITY

AVERAGE DEMAND - DATA POINT

EXAMPLE:

AVERAGE DEMAND = 20 UNITS

DATA POINTS = 8, 28, 13, 7, 15, 25, 17, 33, 40, 9, 11, & 34

*Get the absolute difference

$$20 - 8 = 12$$

SQUARE EACH DIFFERENCE

$$12 \times 12 = 144$$

DIFFERENCES = 12, 8, 7, 13, 5, 5, 3, 13, 20, 11, 9, & 14

144, 64, 49, 169, 25, 25, 9, 169, 400, 121, 81, & 196

AVERAGE OF THE SQUARES

$$= \frac{144 + 64 + 49 + 169 + 25 + 25 + 9 + 169 + 400 + 121 + 81 + 196}{12}$$

$$= 121$$

SQUARE ROOT OF AVERAGE

$$\sqrt{121} = 11 \text{ — STANDARD DEVIATION OF DEMAND}$$

Z-SCORE

=

*Service Factor, based on the
standard deviation of demand*

PROTECTS FROM

1 Z-SCORE = 1 STANDARD DEVIATION OF DEMAND

EXAMPLE:

11 STANDARD DEVIATION OF DEMAND

Z-SCORE = 11

*To protect from 1 standard
deviation of demand you
need 11 units of safety
stock + normal stock*

**HIGHER
Z-SCORE**

= *Less likely you are
to have a stock-out*

1.65 Z-SCORE = **GENERALLY REGARDED
AS ACCEPTABLE EVEN
FOR IMPORTANT STOCK**
*Satisfying demand
with a 95%
confidence level*

*X STANDARD DEVIATION
CHOSEN Z-SCORE*

EXAMPLE:

**NO. OF UNITS
TO STOCK** = **AVERAGE
DEMAND** + **SAFETY
STOCK**

= 20 + (11 X 1.65)

= 20 + 18

= **38**

**MEANS YOU NEED TO STOCK
APPROXIMATELY 38 UNITS**



**LEAD
TIME**

*The time from when you
decide to produce or
order an item until the
time at which the item is
on hand and ready for
sale to the end customer*

FACTORS THAT MAKE IT VARY

**PRODUCTION
DELAYS**

**DELIVERY
DELAYS**

**MATERIAL
DEFECTS**



SYNC STOCK WITH SUPPLY DELIVERY CYCLE

↓
X STANDARD DEVIATION OF DEMAND
SQAURE ROOT OF LEAD TIME

EXAMPLE:

If you calculated standard deviation on a monthly basis, and lead time was 2 months

$$= 11 \times \sqrt{2}$$

$$= 15.56$$

Make sure to convert lead time to the same unit of time measure that you used to determine standard deviation of demand

EXAMPLE:

If you calculated standard deviation on a monthly basis, and lead time was 10 days

$$\begin{aligned}\text{LEAD TIME} &= 10 \text{ DAYS} \times 30.42 \text{ DAYS} \\ &= .329 \text{ month}\end{aligned}$$



$$\text{SAFETY STOCK} = \text{Z-SCORE} \times \sqrt{\text{LEAD TIME} \times \text{STANDARD DEVIATION OF DEMAND}}$$

EXAMPLE:

$$= 1.65 \times \sqrt{2 \times 11}$$

$$= 25.67 \text{ UNITS}$$



IF LEAD TIME IS THE PRIMARY VARIABLE:

$$\text{SAFETY STOCK} = \text{Z-SCORE} \times \text{STANDARD DEVIATION OF LEAD TIME} \times \text{AVERAGE DEMAND}$$

EXAMPLE:

$$= 1.65 \times .43 \times 20$$

$$= 14.3 \text{ UNITS}$$



$$\text{SAFETY STOCK} = Z\text{-SCORE} \times \sqrt{\text{LEAD TIME} \times \text{STANDARD DEVIATION OF DEMAND}^2 + \left[\text{STANDARD DEVIATION OF LEAD TIME}^2 \times \text{AVERAGE DEMAND} \right]^2}$$

$$= 1.65 \times \sqrt{(2 \times 11^2) + (.43 \times 20)^2}$$

$$= 29.3 \text{ UNITS}$$

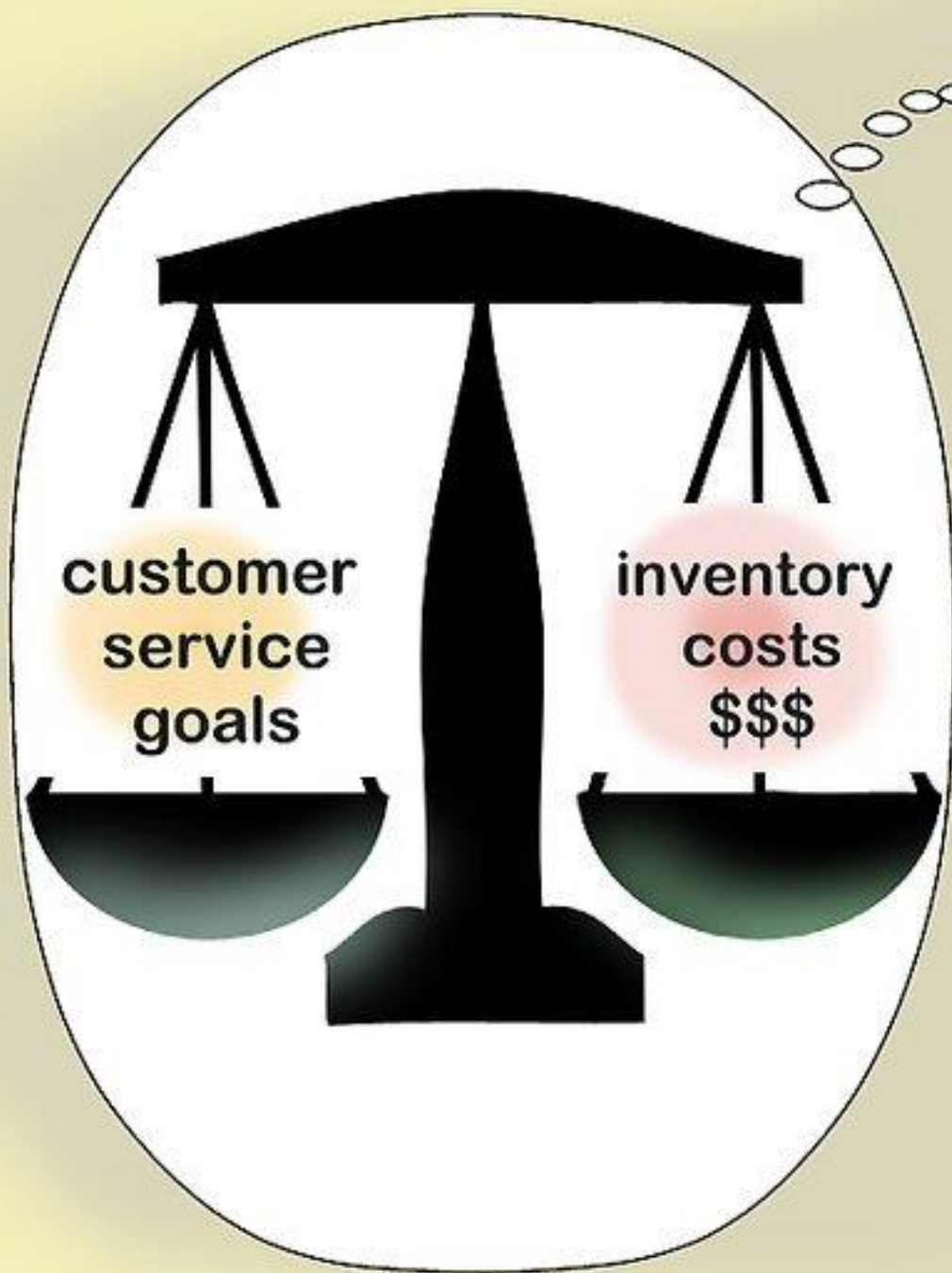


$$\text{SAFETY STOCK} = \left[Z\text{-SCORE} \times \sqrt{\text{LEAD TIME STANDARD DEVIATION OF DEMAND}} \right] + \left[Z\text{-SCORE} \times \text{STANDARD DEVIATION OF LEAD TIME AVERAGE DEMAND} \right]$$

$$= 25.67 + 14.3$$

$$= 13.97 \text{ UNITS}$$





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Satisfying demand with a 95% confidence level

*X STANDARD DEVIATION
CHOSEN Z-SCORE*

EXAMPLE:

NO. OF UNITS TO STOCK = **AVERAGE DEMAND** + **SAFETY STOCK**

= 20 + (11 X 1.65)

= 20 + 18

= **38**

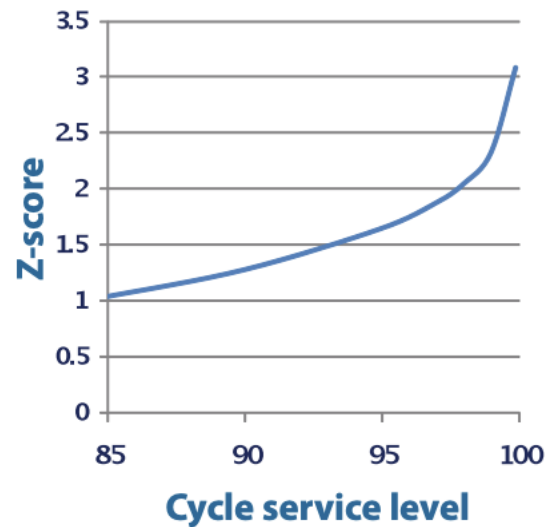
MEANS YOU NEED TO STOCK APPROXIMATELY 38 UNITS

PC = performance cycle, another term for total lead time

T_1 = time increment used for calculating standard deviation of demand

σ_D = standard deviation of demand.

Desired cycle service level	Z-score
84	1
85	1.04
90	1.28
95	1.65
97	1.88
98	2.05
99	2.33
99.9	3.09



— FIGURE 2: Relationship between desired service level and Z-score —

https://web.mit.edu/2.810/www/files/readings/King_SafetyStock.pdf

$$\text{Safety stock} = Z \times \sqrt{\frac{PC}{T_1}} \times \sigma_D$$

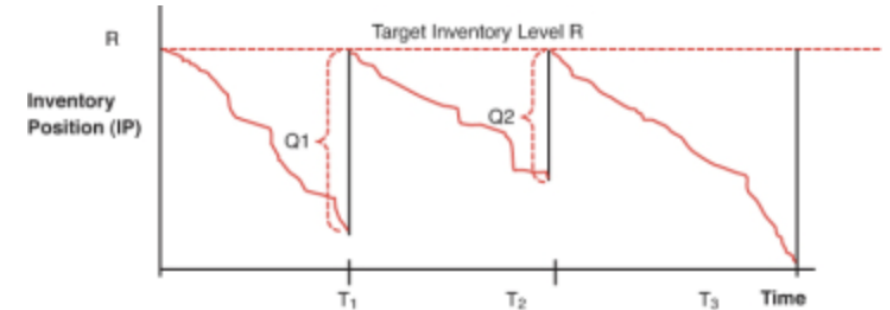


Figure 1-6 Fixed-time period system

$$Q = R - IP$$

where:

Q = order quantity

R = target inventory level

IP = inventory position

<https://www.informit.com/articles/article.aspx?p=2167438&seqNum=7>

Fixed-Order Quantity System

The first policy choice is through a system called a *fixed-order quantity system*. Therefore, two variables define this system and answer the two basic questions of when to order and how much. The first is an order quantity, Q , and the second is a reorder point, ROP . As the name suggests, the quantity ordered with this system is constant or fixed and is denoted by Q . An order is placed when the inventory position drops to some predetermined level. This predetermined level is called the reorder point and is usually noted as ROP . Together these variables specify when to place an order: *when inventory reaches the ROP*. They also specify how much to order: *the quantity Q* .

A graphical presentation of this model is shown in [Figure 1-5](#). Notice that the system assumes a constant demand rate of d by which the inventory position (IP) is reduced. When the IP reaches the ROP , an order is placed for the quantity Q . When goods arrive, the inventory is replenished, and all at once the inventory position is increased by Q . However, inventory cannot arrive the moment an order is placed as there is a certain amount of lead time, L , during which we have to wait for the order.

In the classic version of this system Q is computed as the economic order quantity (EOQ)—an economically optimal order quantity—which we will compute later. For this reason this system is sometimes called the *economic order quantity (EOQ) model*. Other terms used to describe it are the *Q -model* as the quantity Q is constant. Sometimes it is called a *continuous review system*, as the inventory levels are continuously monitored. The model has even been called a *sawtooth model*, as the graph of inventory looks like a sawtooth. All these terms refer to the same type of inventory system, describing different features of the system itself.

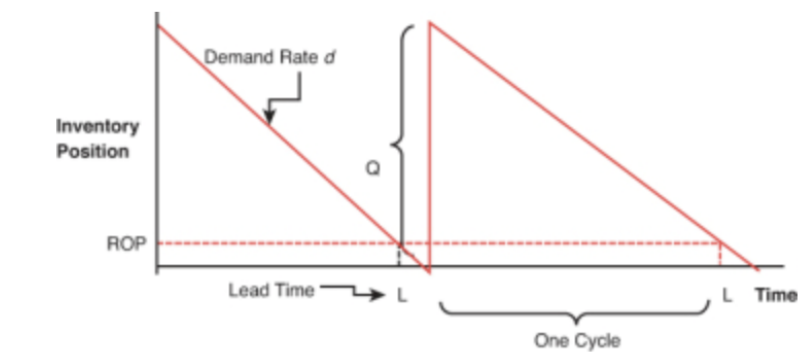


Figure 1-5 Fixed-order quantity systemt

Here inventory is monitored on a continual basis, and the assumption is that we always know the current level of inventory. When inventory levels reach the ROP , an order of quantity Q is placed. For example, let's assume that a grocer uses a fixed-order quantity system for its inventory of canned tomato soup. Its policy may be that it always orders 200 cans of soup, for which it gets a nice quantity discount, and the order is made when the number of cans of soup drops to 25. Therefore, the $ROP = 25$, and the $Q = 200$.

Fixed-Time Period System

The second inventory policy is determined by a system called a *fixed-time period system* shown in Figure 1-6. This system checks inventory levels in fixed time intervals labeled as T . The result is that the quantity ordered varies based upon the inventory position when the system is checked. The system sets a target inventory level it wants to maintain, say R . Inventory is checked every T intervals, say every week or every two weeks, and an order is placed to restore the inventory level back to R . Based on the inventory level at time period T , the amount of inventory that needs to be ordered will be some quantity Q that varies from period to period. This quantity Q is the difference between the target inventory R and how much inventory is in stock—the IP at time T :

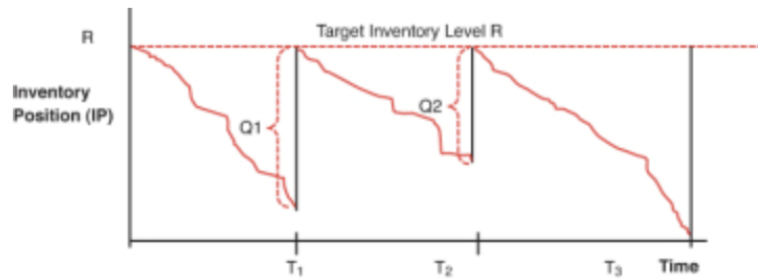


Figure 1-6 Fixed-time period system

$$Q = R - IP$$

where:

Q = order quantity

R = target inventory level

IP = inventory position

Two variables define this system and answer the two basic questions of when to order and how much: T and Q . They specify when to place an order: *at time interval T* . They also specify how much to order: *quantity Q , computed as the difference between the target inventory, R , and the inventory position, IP* . Sometimes this system is called the *periodic review system* to indicate that the inventory level is checked periodically, rather than continuously.

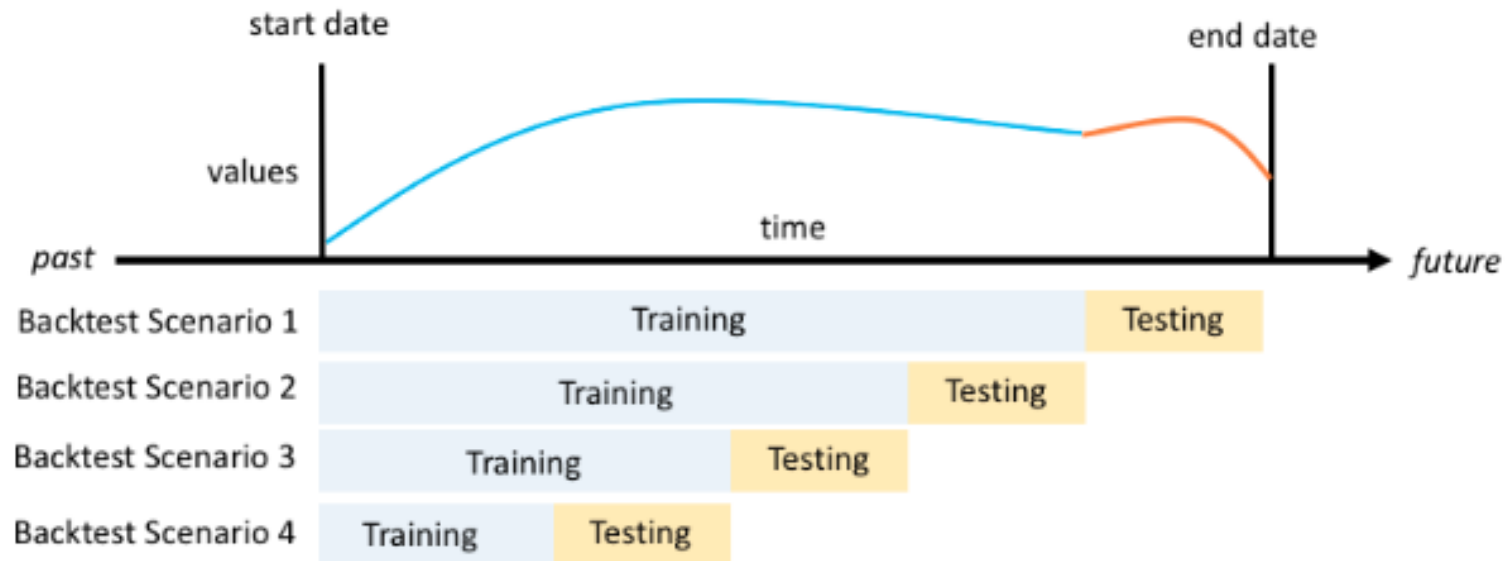
We can assume that an auto manufacturer uses a fixed-time period mode for its inventory of alternators. Also, let's say its policy is to check inventory levels every two weeks, and that it has a target inventory level $R = 5,000$ alternators. If after two weeks the company checks its inventory level and finds its inventory position $IP = 2,800$ alternators, it would place an order for quantity $Q = R - IP = 5,000 - 2,800 = 2,200$ alternators. This is essentially how this policy system works.

The biggest difference between the fixed-order quantity system and the fixed-time period systems is in the timing and quantities of the orders placed. With the fixed-order quantity system inventory is checked on a continual basis and the system is prepared to place orders multiple times per year on a random basis. This has an advantage of providing greater system responsiveness, but it also requires administrative processes to be in place on a continual basis. In addition, as different inventory items reach their reorder points at different time periods, it might be difficult to obtain quantity discounts based on a bundled order.

The fixed-period order system requires carrying more safety stock inventory. The reason is that with this system we do not check the IP on a regular basis, and a sudden surge of demand could lead to a stockout. This system, however, allows more organized purchasing as inventory levels are checked in set time intervals. Orders can be bundled and quantity discounts obtained more easily, which can provide an advantage. In a fixed-order quantity system different items may reach reorder points at different times generating many orders at random intervals. On the other hand, a fixed-period system could ensure that inventory levels are checked on a regular basis for all items—say every two weeks. Then the orders for all the items could be bundled.

<https://www.informit.com/articles/article.aspx?p=2167438&seqNum=7>

The backtest window must be at least as large as the forecast horizon, and smaller than half the length of the entire target time-series dataset. You can choose from between 1 and 5 backtests.



Safety Stock Calculation

The safety stock depends on the service level that you specified in the MRP II view of the material master record and on the accuracy of the forecast. The more accurate the forecast, the smaller your safety stock can be.

The following figure shows that, without safety stock, customer demand can be satisfied by 50%. It also shows that it is almost impossible to satisfy customer demand 100% of the time. Factor R describes the relationship between forecast accuracy and service level (SL).

If replenishment lead time is greater than the forecast period by factor W, then the mean absolute deviation (MAD) is recalculated for this period (formula 17).

If the material is produced in-house, the delivery time is: opening period + in-house production time + goods receipt processing time. It is expressed in workdays. The forecast period is taken from the material master record and is also expressed in workdays.

If the material is procured externally, the delivery time is: goods receipt processing time + planned delivery time + goods receipt processing time. It is expressed in calendar days. The forecast period is taken from the material master record and is also expressed in calendar days.

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The safety stock calculation may lead to rounding effects.

Minimum Safety Stock

You can specify a minimum safety stock. If the result of the safety stock calculation is lower than this limit, the safety stock is then automatically set to this minimum value. You enter the minimum safety stock in the material master record (the MRP 2 screen).

<https://help.sap.com/docs/SCMCSSSS/1d3a3ab8bcca43c79a75916ea2e51f60/c8932451730e10149785883a656123e7.html>

Service Level Calculation

After the service levels have been optimized, the service level must be calculated in the next step.

In the selected group, the service levels are initially selected so as to achieve the optimized service level. This is done using the following formula, which must be performed for each material.

Formula for Optimized Service Level

Optimized service level = (forecast value of the current period + MAD of the current period) * (optimized service level as a percentage/100)

Example

Forecast value for March 2012 in the base units of measure: 1000

MAD: 50

SL as a percentage: 96 $(1000+50) * (96/100) = 1008$

The safety stock that the above service level would enable must now be determined using the relevant safety stock method. For this purpose, the level of the safety stock is first determined using the following formula:

Formula for Optimized Safety Stock Quantity

Optimized safety stock quantity = optimized service level – forecast value of the current period

Example

$1008 - 1000 = 8$

The result of the SL-optimization shows you various information from the material master as well as calculated values from the **safety stock simulation**:

SL-optimized safety stock simulation																											
Number of selected Materials														Total Service Level													
Number of updated Materials														94,9													
Number of not updated Materials														562													
Material	Plant	ABC	XYZ	LM	EFG	UVW	Cycle	StOrg	MRPC	MT	Mat Group	Proc. type	Typ U	/ Crd	Rel. price	Cat	RLT C	SL lev	Max. SL	AchTrgt SL	Min. SL	SSe OptE	SSV DEM	ROP OptERP	ROPV OptE	SL opt.	Updated
CS11_1_FM_SALSO001	0001	C	X	N	E	W	D	0001	002	FERT 001	E	VV	PC	0	EUR	0,00	4	63,3	99,9	96,0	99,0	20,622	0,00	20,622	0,00	✓	
CS_FMO_FERT_F_0001	1000	C	X	N	E	W	D	1000	000	ZFER	F	PD	PC	0	EUR	0,00	8	0,0	99,9	96,0	99,0	20,912	0,00	20,912	0,00	✓	
CS_FMO_FERT_F_0002	1000	C	X	N	E	W	D	1000	001	ZFER	F	PD	PC	0	EUR	0,00	8	0,0	99,9	96,0	99,0	0,015	0,00	0,015	0,00	✓	
CS_MRP_FERT_F_0003	1000	C	X	N	E	V	D	1000	001	FERT 01	E	PD	PC	1	EUR	4,00	14	0,0	99,9	90,0	99,0	296,225	824,90	296,225	824,90	✓	
CS_MIGO_F_00000001	1000	C	X	N	E	V	D	1000	001	ROH 01	F	PD	PC	1	EUR	4,00	7	65,5	99,9	90,0	99,0	0,006	0,02	0,006	0,02	✓	
CS_SSS_FERT_F_0005	1000	C	X	N	E	V	D	1000	001	FERT 01	E	PD	PC	1	EUR	4,00	14	70,5	99,9	96,0	99,0	618,940	2.475,76	618,940	2.475,76	✓	
CS_SSS_FERT_F_0002	1000	C	X	N	E	V	D	1000	001	FERT 01	E	PD	PC	1	EUR	4,00	14	68,4	99,9	96,0	99,0	550,676	2.202,70	550,676	2.202,70	✓	
CS_ROH_F_00000002	1000	C	X	N	E	V	D	1000	001	ROH 01	F	PD	PC	1	EUR	4,00	7	61,8	99,9	96,0	99,0	0,006	0,02	0,006	0,02	✓	
CS_ROH_F_00000001	1000	C	X	N	E	V	D	1000	001	ROH 01	F	PD	PC	1	EUR	4,00	7	61,8	99,9	96,0	99,0	0,006	0,02	0,006	0,02	✓	

Result View for SL-Optimization

The columns with a green background in the results view show the calculated safety stocks and orders point in terms of both quantities and values. These results are updated when a material master update is required.

Forecast values play an important role in SL-optimization. If neither forecast values nor an average consumption can be determined for a material, SL-optimization cannot be performed for this material. This is shown by the missing checkmark in the **SL opt. (optimized)** column on the one hand and by N/A (Not Available) in the **Forecast source** column on the other.

In addition, the difference between the optimized target service level (column **AchTrgt SL**) and the target service level entered (column **AltTrgt SL**) enables you to recognize a material for which SL optimization has been performed successfully. If both values are identical, the material has been assigned the entered (non-optimized) target service level. The reorder point and safety stock are then calculated based on this service level.